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LESSER: Post-Training Data Selection with Output-Layer Gradients

Output-Layer Gradients Are a 9.7× Cheaper Proxy for Full-Gradient Data Selection While Preserving Batch-Level Alignment and Downstream Performance

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Gradient-based data selection for LLM post-training requires an expensive full-backward pass for every candidate sample; LESSER shows that restricting gradient computation to the output layer reduces cost by up to 9.7× while matching full-gradient performance, because batches selected by both methods remain well-aligned with the validation gradient even when individual sample rankings differ.

AT A GLANCE

Problem: Full-parameter gradient computation for post-training data selection is intractable at scale.

Why it matters: The right training data can substantially improve downstream performance; efficient selection unlocks practical use.

Method: Restrict gradient computation to the output (language-model head) layer only; use as a drop-in wrapper for any selection algorithm.

Result: 9.7× FLOP reduction for SFT and 3.0× for RL selection, with no downstream performance loss.

Evidence: Evaluated across multiple gradient-based selection methods on SFT and RL post-training benchmarks.

The Big Picture

The choice of which data to use for post-training an LLM substantially shapes its downstream capabilities. Gradient-based data selection formalizes this choice: rank candidate data points by how well their per-sample gradients align with the gradient on a small, high-quality validation set, then train on the top-ranked examples. The method is principled and effective but computationally prohibitive at scale—each candidate requires a full backward pass through a billion-parameter model.

LESSER (Last-layer Efficient Sample Selection via

Output-layer gRadients) asks: does the output layer’s gradient contain enough information to perform effective selection, at a fraction of the cost of backpropagating through the entire network?

Why Existing Approaches Fall Short

Full-gradient cost is prohibitive. For a model with N parameters and a candidate pool of M samples, gradient-based selection requires $O(NM)$ work. At $N = 7 \times 10^9$ and $M = 10^5$, this is impractical without approximation.

Embedding-layer proxies introduce extra hyperparameters. LESS and related methods use LoRA projections of embedding-layer gradients, introducing a rank hyperparameter and a random-projection step that adds engineering overhead.

Sample-level alignment is not the operative quantity. Prior work implicitly assumes that a proxy must rank individual samples identically to full gradients. LESSER demonstrates that batch-level alignment—the quantity that governs gradient descent quality—is preserved even when sample rankings differ.

Core Mechanism

LESSER restricts gradient computation to the **output layer** (language model head)—the final linear projection W_{head} from hidden states to vocabulary logits. This layer has far fewer parameters than the full model but its gradient directly reflects the training objective.

The output-layer gradient for a data point x is:

$$\nabla_{W_{\text{head}}} \ell(x, \theta) = \frac{\partial \ell}{\partial \text{logits}} \cdot h_x^\top$$

where h_x is the final hidden state and $\frac{\partial \ell}{\partial \text{logits}}$ is the softmax loss gradient. This requires only a single matrix multiplication and does not backpropagate through transformer layers.

Technical Formulation

Standard gradient-based selection scores candidate x as:

$$s(x) = \langle \nabla_{\theta} \ell(x, \theta), \nabla_{\theta} \mathcal{L}(\mathcal{V}, \theta) \rangle$$

LESSER replaces θ with the output-layer parameters θ_L :

$$s_{\text{LESSER}}(x) = \langle \nabla_{\theta_L} \ell(x, \theta), \nabla_{\theta_L} \mathcal{L}(\mathcal{V}, \theta) \rangle$$

The key empirical finding—that batch-level alignment is preserved—can be stated as follows. Let B_{full} and B_{LESSER} be the top- k batches selected by each method. Then empirically:

$$\langle \nabla_{\theta} \mathcal{L}(B_{\text{LESSER}}, \theta), \nabla_{\theta} \mathcal{L}(\mathcal{V}, \theta) \rangle \approx \langle \nabla_{\theta} \mathcal{L}(B_{\text{full}}, \theta), \nabla_{\theta} \mathcal{L}(\mathcal{V}, \theta) \rangle$$

The intuition is that the last layer aggregates information from all earlier layers, so its gradient captures the dominant component of the full gradient along the data manifold.

Learning or Inference Procedure

1. **Cache validation gradient:** Run a forward-backward pass on the validation set using only output-layer parameters. Cache $g_V = \nabla_{\theta_L} \mathcal{L}(\mathcal{V}, \theta)$.
2. **Score candidates:** For each candidate x_i , run a forward pass; compute $g_i = \nabla_{\theta_L} \ell(x_i, \theta)$ via one matrix multiplication; score $s_i = \langle g_i, g_V \rangle$.
3. **Select top- k :** Rank candidates by s_i ; select the top k .
4. **Post-train:** Fine-tune on selected data with standard SFT or RL objective.

What the Guarantee Says

There is no formal guarantee that output-layer ranking preserves full-gradient ranking at the sample level—and indeed it does not, in general. The operative guarantee is empirical and at the batch level: across all tested settings, batches selected by LESSER achieve gradient alignment with the validation set indistinguishable from those selected by full-gradient methods, and deliver matching downstream task performance.

Experimental Findings

- **FLOP reduction:** LESSER reduces feature-extraction cost by **9.7×** for SFT and **3.0×** for RL benchmarks.
- **Performance parity:** On downstream evaluations, LESSER matches full-gradient selection within measurement noise across all settings.
- **Sample vs. batch alignment:** Output-layer and full-gradient Spearman sample-ranking correlations vary widely, but batch-level gradient alignment with the validation set is consistently preserved.
- **Compatibility:** LESSER wraps LESS, DataInf, and other selection algorithms without modifying the selection logic.

Ablations and Interpretation

Layer depth sweep: Ablations over which layer’s gradients to use show a monotone improvement with depth; the output layer is best. Mid-layer gradients reduce cost but widen the quality gap.

Validation set size: Performance degrades gracefully as validation size shrinks; 128 examples suffice for reliable selection.

SFT vs. RL gap: The FLOP saving is larger for SFT (9.7×) than RL (3.0×) because RL requires gradients over longer generated sequences, making each full backward pass more expensive and the relative saving smaller.

Reference

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